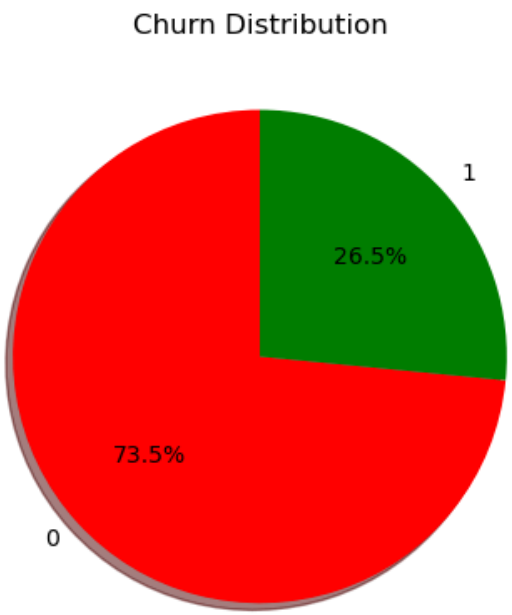
Customer Churn Analysis & Prediction

Introduction

This project analyzes customer churn behavior within a telecommunications dataset to identify high risk customers who are likely to discontinue service as well as the key factors influencing such behaviors. The aim of this project is to extract actionable insights from the customers data that can inform strategies aimed at customer retention.

Data & Methodology

The dataset used consists of records of customers who left the telco within the last month. The target for this dataset is the “churn” column which holds the data of the individuals who discontinued the service of the telco. The dataset has a total of 21 columns and 7,043 records. The data has some major features such as the senior citizen, gender, tenure, partners, and contract. An initial analysis of the dataset indicates an imbalance based on the senior feature and churn label, with non-senior customers and non-churn representing the majority. This imbalance is important to consider as it might influence the interpretation of results. The Pie Chart in fig. 1 shows the percentage rate of seniors to non-seniors and churn to non-churn.



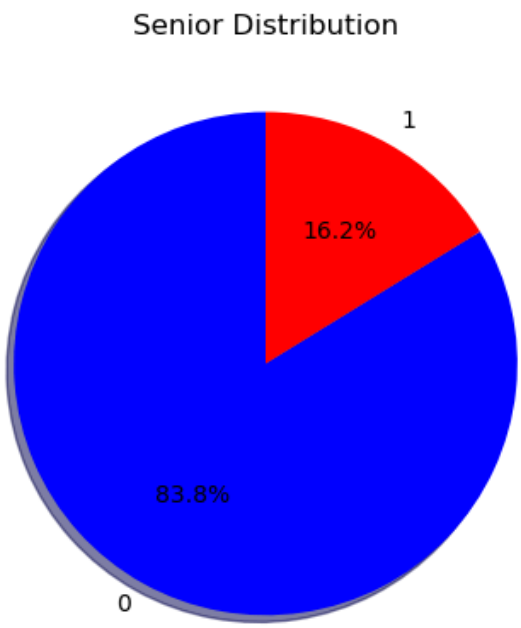


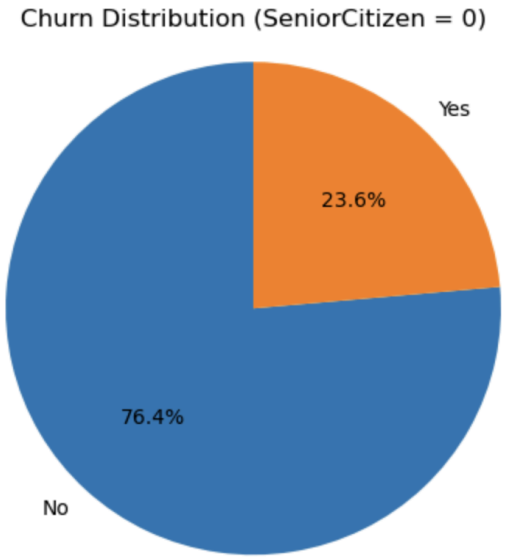
Fig. 1

Exploratory Data Analysis (EDA)

The first step in this project was the data cleaning and preprocessing which indicated the dataset had no missing values, no duplicates and the majority of the data types were correct except the total charges feature which was changed from an object to an integer type. Following this, exploratory data analysis was conducted to examine relationships between customer characteristics and churn behavior. The analysis was broken down into various groups to determine key relationships and these features below showed stronger results:

- Senior Citizens vs Churn
- Senior Citizens vs Partner vs Gender vs Dependents vs Churn
- Senior Citizens vs Contract vs Partner vs Churn
- Senior Citizens vs Internet Service vs Churn

The Senior Citizens vs Churn insight shows that 23% of non-senior citizens and 41% of senior citizens churned last month. This shows a significant number of senior customers churned last month and this can be attributed to factors such as affordability, usability or different customer service needs. Fig. 2 below shows that.



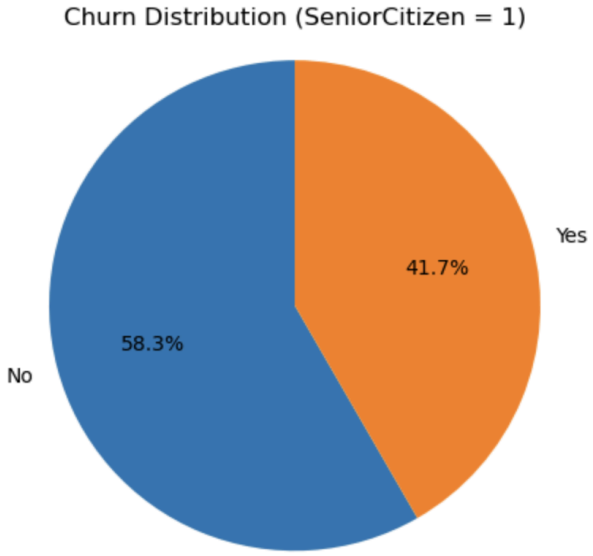


Fig. 2

The analysis subsequently focuses on customers classified as senior citizens in order to understand the contributing factors to their high churn rate. Within this segment additional features namely: gender, partner and dependents were examined against the churn label to determine their potential influence on customer retention.

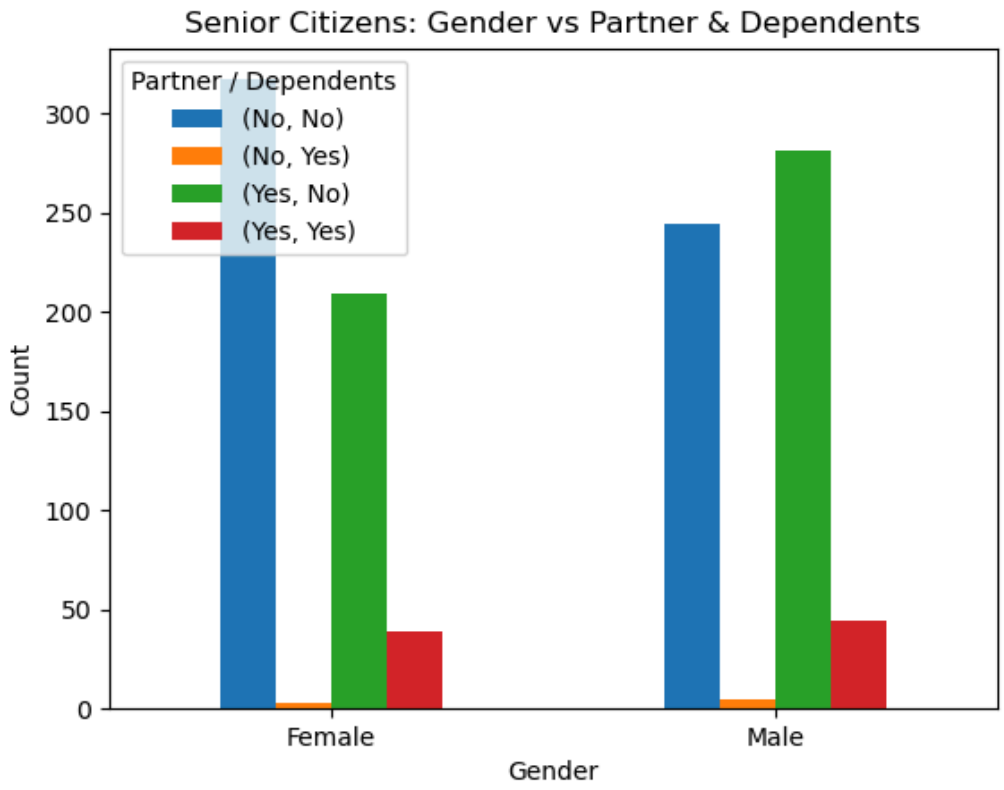
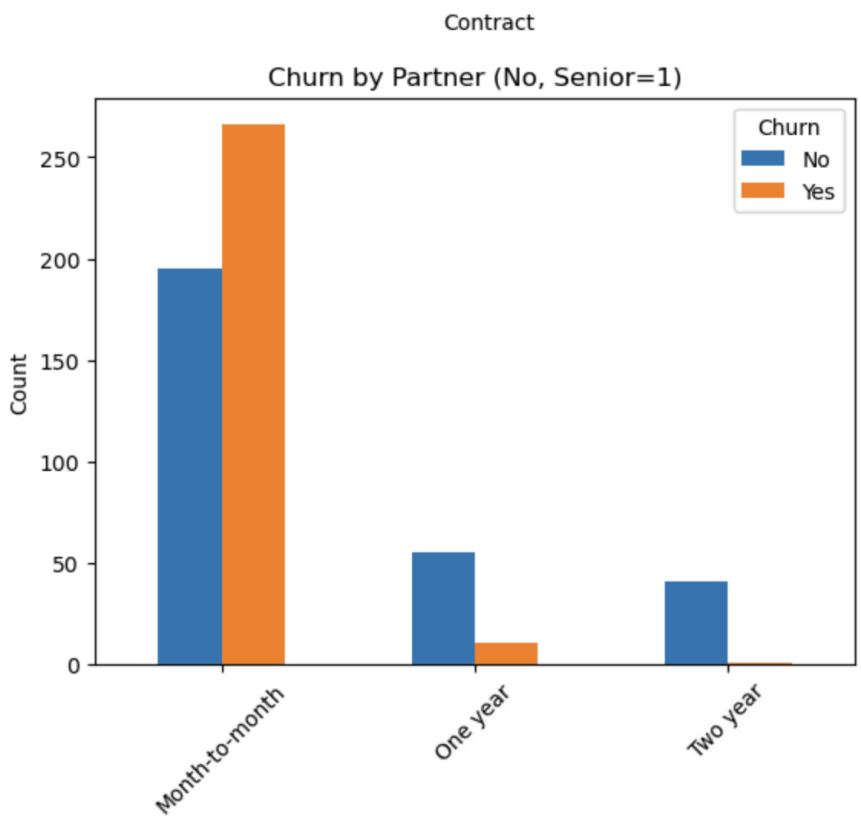
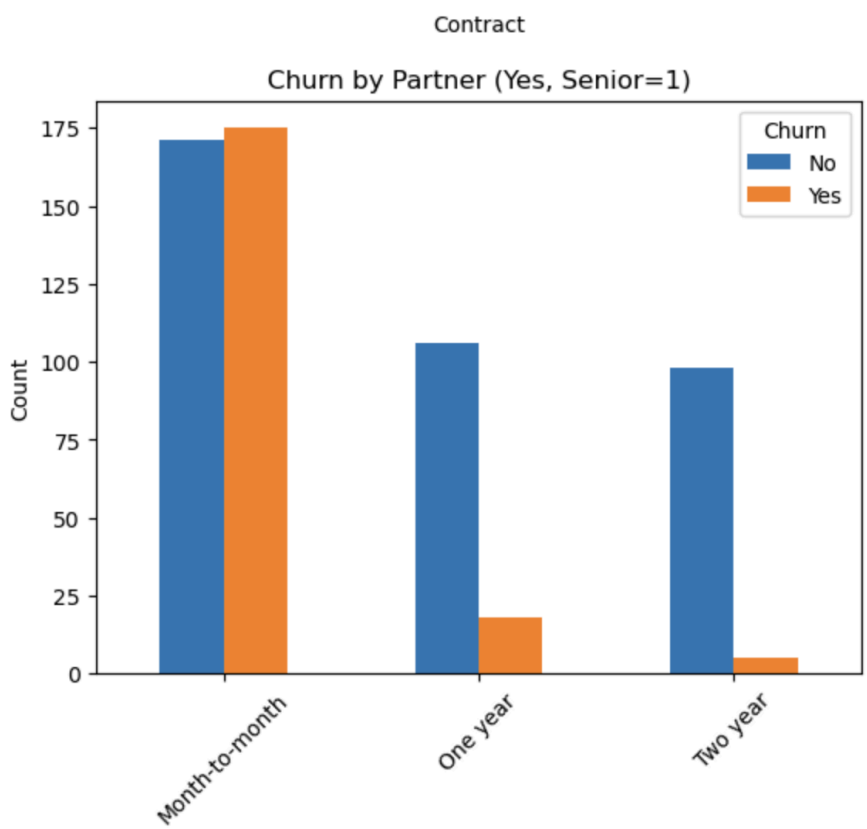


Fig. 3

The result as shown in fig.3 above shows that senior citizens without both dependents and partners as well as those with partners but no dependent constitute the highest group of customers that churned within the last month. This indicates that household composition might play a vital role where reduced familial or dependent could be associated with a higher likelihood of service discontinuation. However, this relationship should be interpreted cautiously, as further analysis would be required to establish definite cause.

Further analysis into the senior citizens, contract and partner columns against the churn label shows that the majority of customers are subscribed on a monthly contract which accounts for 55% of the entire customer base. Notably, this contract category accounts for the highest number of customers representing 23.5% discontinuing this service. This finding indicates flexibility as a potential driver of churn as short term commitments may be an indicator for customers leaving. Fig. 4 below shows that.


Contract
Churn by Partner (No, Senior=1)
Churn
No
Yes
Count
0
50
100
150
200
250
Month-to-month
One year
Two year


Contract
Churn by Partner (Yes, Senior=1)
Churn
No
Yes
Count
0
25
50
75
100
125
150
175
Month-to-month
One year
Two year

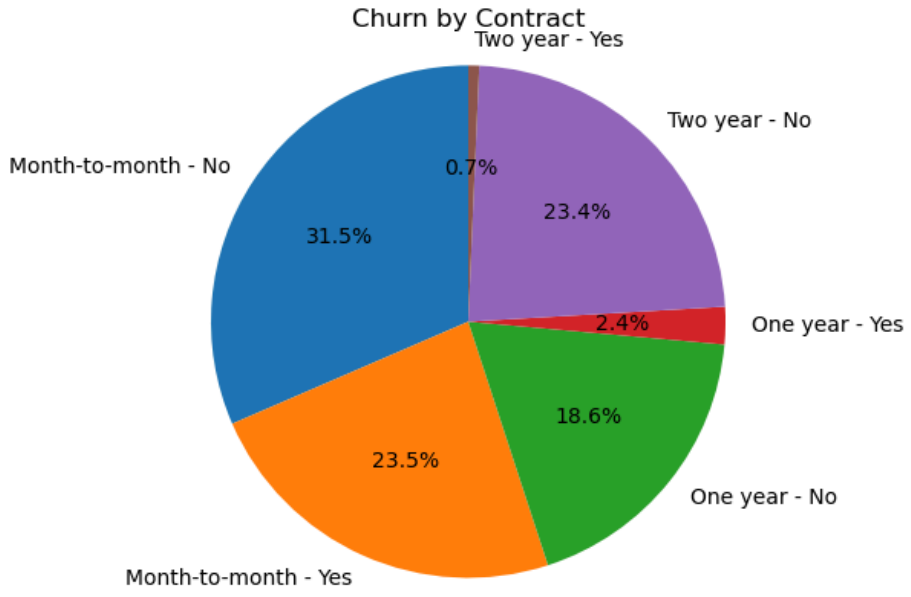


Fig. 4

In addition, the analysis also shows that the majority of the customers used the Fibre Optic internet service. Among these customers 18.4% churned within the last month, representing the largest share of churned customers by service type. This may point to underlying factors such as pricing, service reliability, or other customer expectations associated with fibre optic offerings.

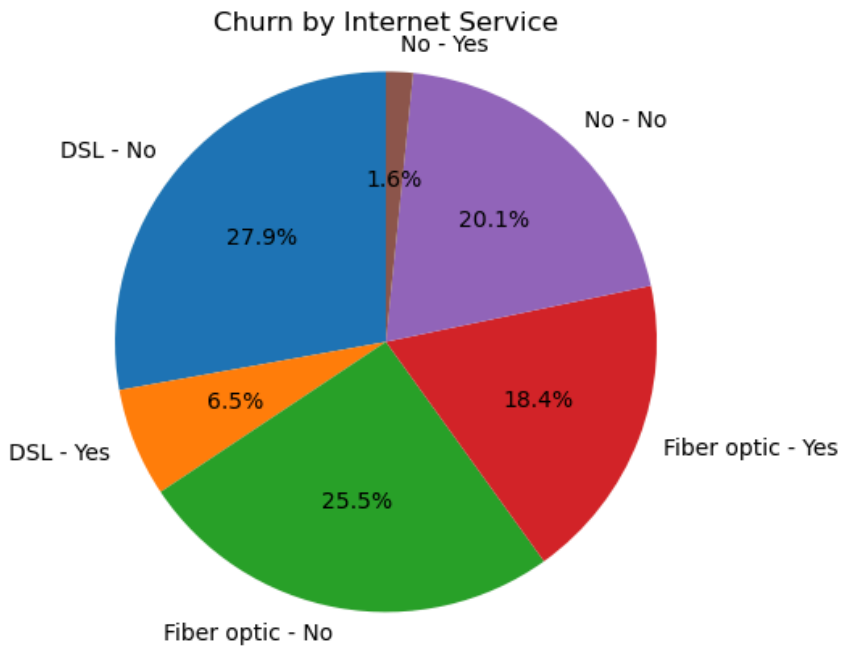


Fig.5

Key Findings

Following the completion of the exploratory data analysis, categorical features were transformed into numerical values to perform a correlation analysis. A correlation matrix was computed to examine the relationships between features and their association with the target variable ("Churn"). As illustrated in fig 6 the correlation heatmap highlights both positive and negative relationships among the variables.

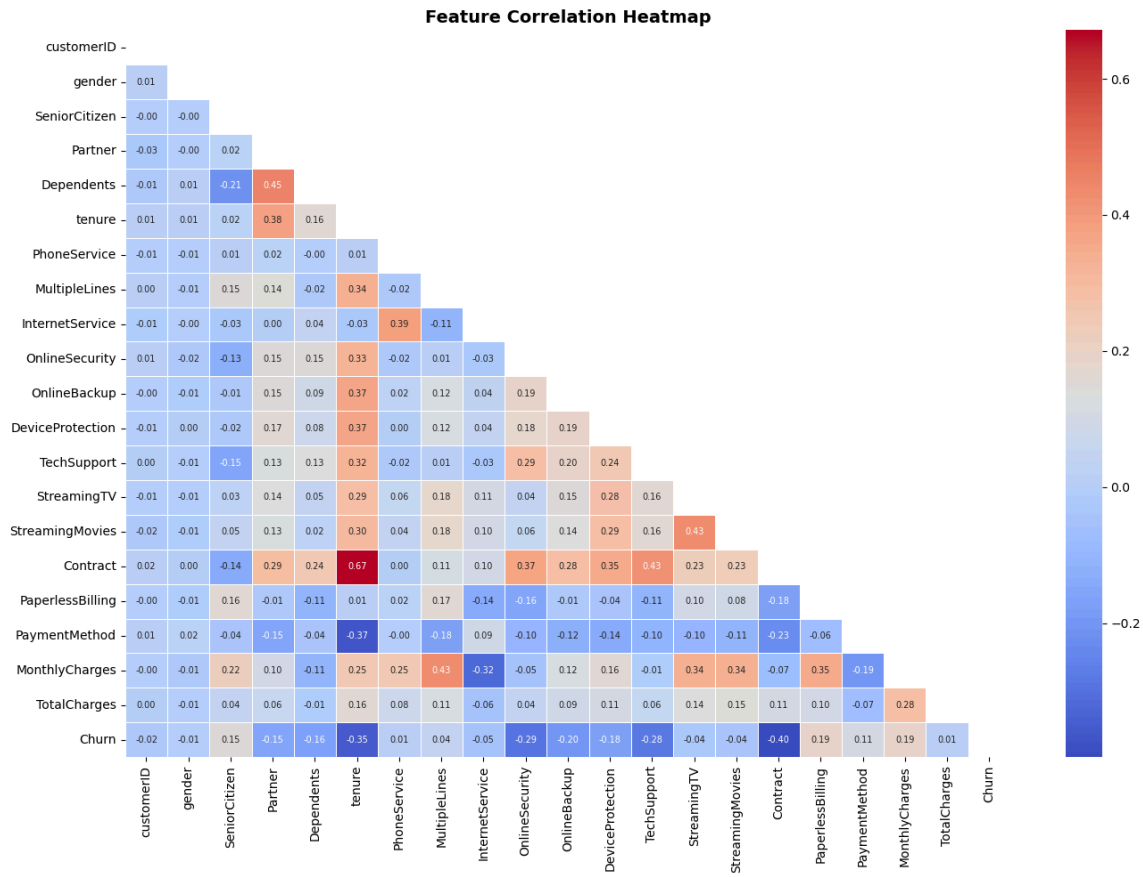


Fig. 6

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The analysis reveals a strong positive correlation between tenure and total_charges = 0.85. This indicates that as customer tenure increases, total charges also increase suggesting that both features capture similar information resulting in highly correlated features = duplicate signals.

To address this issue, the total charges feature was dropped to mitigate potential model overfitting and reduce multicollinearity. The remaining features were then used to train a random forest model to evaluate feature importance. Fig.7 shows the importance ratios.

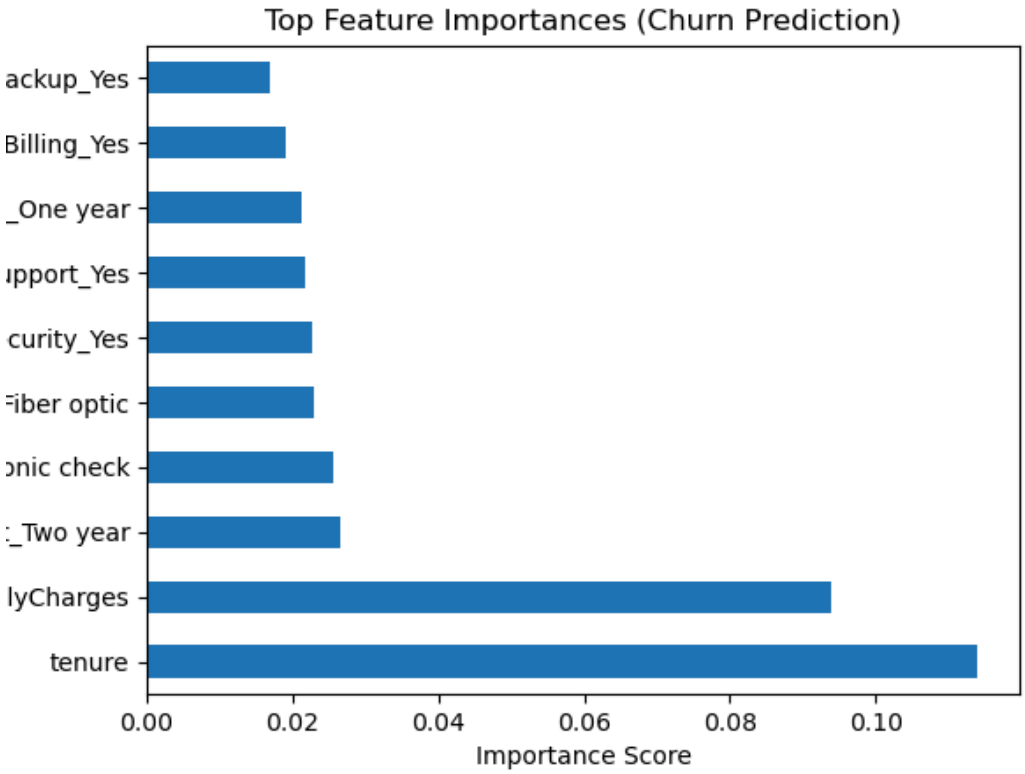

Fig. 7

Model Prediction

The dataset was split into training and testing using a 70/30 ratio. Stratified sampling was applied to ensure label division consistency across both datasets. The dataset was initially trained on a random forest model and it achieved 0.79 accuracy, with a recall of 0.49, precision of 0.64 and an f1 score of 0.56. Following the result, a confusion matrix was plotted to show how well the model performed on the churn vs non-churn label. The results indicated that, out of 2,113 observations in the test set, the model correctly classified 1,400 non-churn instances and 276 churn instances, while having 285 false negatives and 152 false positives demonstrating a stronger performance in identifying non-churn customers compared to churn customers. These results also show that a significant portion of churned customers are not being identified indicating the model is biased towards predicting non-churn outcomes, a common issue in imbalanced datasets. This limits the model's effectiveness in proactive customer retention.

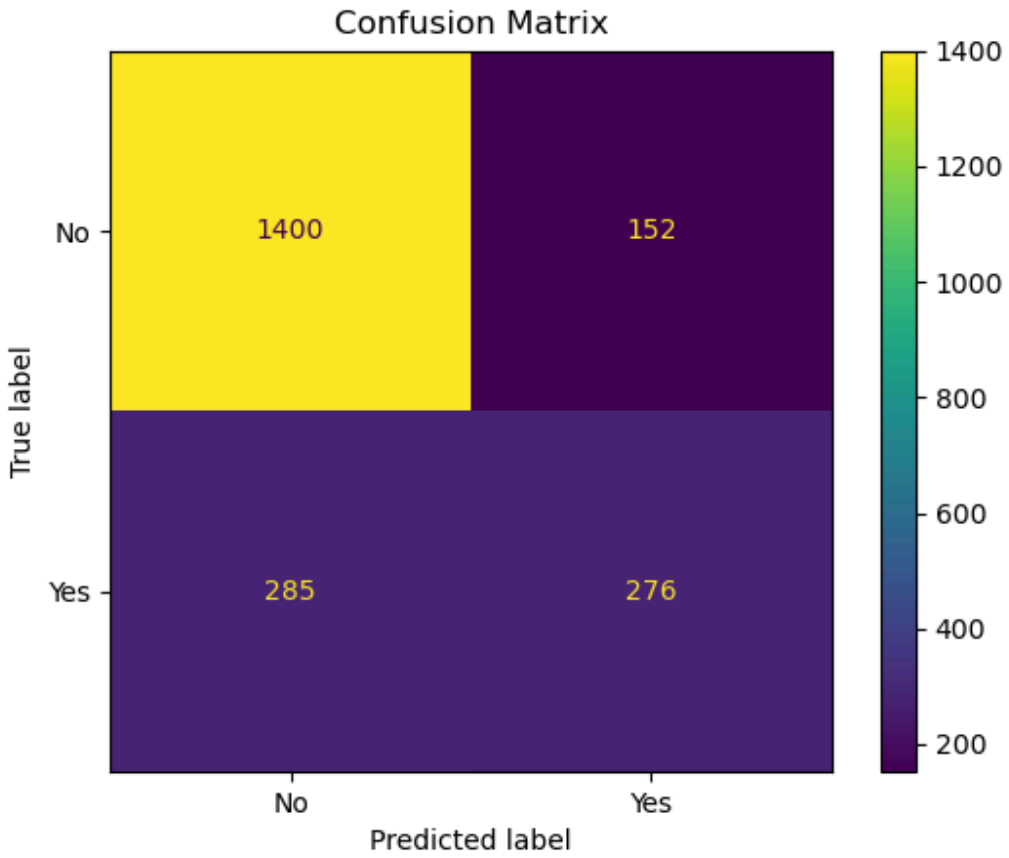

Fig. 8

Subsequently, feature selection was performed using the top ten (10) important features derived from the model and this resulted in a slight decrease in accuracy to 0.78, with a recall fixed at 0.49, a decrease in precision and f1 score to 0.58 and 0.53 respectively. Despite a significant reduction in feature dimensionality, this drop in precision suggests that important predictive signals may have been lost, indicating that the full feature set provides better model performance. overall.

To further validate model robustness, additional experiments were implemented using cross validation, Logistic model, and XGBoost model. However, the cross validation and XGBoost model did not yield significant performance improvements over the Random Forest model but the Logistic model using the full features and feature selection method outperformed the Random Forest models. The Logistic model resulted in an accuracy score of 0.76, with a recall of 0.76, precision of 0.53 and an f1 score of 0.63 outperforming the Random forest model in identifying churned customers as shown in fig 9.

Model	F1	Accuracy	Recall	Precision
Random Forest	0.56	0.79	0.49	0.64
Logistic	0.63	0.76	0.76	0.53
Logistic (tuned)	0.61	0.72	0.82	0.49
XGBoost	0.54	0.77	0.49	0.59

Fig 9.

Furthermore, threshold tuning was applied to the Logistic model to optimize performance. A lower threshold of 0.4 improved recall 0.82, enabling the model to identify a higher proportion of churned customers albeit reducing precision and overall accuracy. At this point both models are great as they were not evaluated using accuracy alone but other factors such as the recall, f1 score and precision. They both support the objective of this project as model selection should be guided by business priorities and balancing the trade between maximizing churn detection and reducing false positives.

Business Recommendations

Based on the findings of these analyses, several targeted strategies can be implemented to reduce customer churn.

- Introduce Incentives for Long-Term Contracts
  Customers on a month to month contract account for the majority of churned rates. This indicates other companies might be offering better contracts and benefits. Offering incentives such as loyalty programmes, discounts, better benefits may encourage a shift in customers to longer term commitments and reduce attrition.

- Develop Targeted Retention Strategies for Senior Customers
  41.7% of senior citizens constitute the majority of churned rates. This implies that an adjustment to the service offerings for senior citizens needs to be implemented and tailored to their specific needs such as simplified plans, improved customer service.

- Evaluate Fibre Optic Service Offering
  Fibre optic users account for the highest rate of churned customers. This indicates potential issues relating to either the pricing or service offerings of fibre optic internet service. This should be further investigated and addressed.

- Enhance Customer Engagement and Loyalty Programs

Improving customer engagement through personalized communication and loyalty incentive programmes can help retain customers.

Conclusion

This analysis highlights several key findings regarding customer churn within the telecommunications dataset. Firstly, the dataset itself indicates an imbalance with the non senior citizen comprising 83.8% and senior citizens accounting for 16.2%. Despite this disparity, senior citizens demonstrate a higher churn rate (41.7%) compared to non-senior citizens (23.6%) which indicates age-related factors may play an important role in customer retention.

Additionally, the analysis revealed that customers who selected the fibre optic internet service accounted for the highest group of churned customers representing 18.4% of the dataset. This suggests that factors such as pricing, service reliability may influence customer dissatisfaction.

Furthermore, the contract types feature emerged as a strong determinant. The month to month group of customers accounting for 23.5% exhibited the highest rate of churned customers. This reinforces that shorter term contracts allows for flexibility and multiple options which might negatively affect the telco company.

Finally, the findings indicate that senior citizen, contract type, and internet service type are key factors associated with customer churn. These insights provide a foundation for developing targeted retention strategies aimed at reducing customer attrition and improving long-term customer value.